



Investigation report

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads -
AAIB-style causal and contributory factors report

Drafted 30/09/2026

INVESTIGATION WORKFLOW SUITE

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Synopsis

On 07/07/2026 the MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers, grounded on Boolarong Bank at Kanooka Heads. It was inbound through the Weerona Channel on service TF18, in darkness and on a falling tide, with a master, a deckhand, a kiosk attendant and 38 passengers on board. The chart plotter display froze at 18:07:06 and did not recover. The transit continued without any reduction of speed. The vessel's heading increased from 195 to 200 degrees and was then held, and the vessel set west out of the 40 m wide dredged channel while the master searched for the No. 4 beacon. The echo sounder's shallow alarm sounded 16 seconds before contact. The vessel struck the bank at 6.8 kt at 18:12:04 and came to rest 74 m west of the charted centreline and 54 m outside the channel. The master stopped both engines and made no attempt to refloat, a decision which very likely reduced the risk to those on board.

The starboard hull plating was split in way of the No. 2 void, which flooded progressively. The starboard propeller, shaft and rudder stock were damaged and the starboard skeg was torn away. There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. All passengers were landed by the port workboat by 19:26. The vessel refloated on the tide at 00:20 and was withdrawn from service, and the channel was closed to commercial traffic for 19 hours.

The investigation found three causal factors:

- The vessel left the dredged channel and tracked onto the bank, which was virtually certain to have contributed.
- The plotter freeze very likely occurred and was extremely likely to have contributed.
- Continuing the transit at eight knots after the freeze, rather than stopping in the channel, was extremely likely to have contributed.

Several contributory factors were also identified:

- The winter timetable moved TF18 into darkness without reopening the route risk assessment, which restricted the channel to daylight transits. This likely contributed.
- There was no night passage plan and no hold point if the primary means of fixing was lost. This very likely contributed.
- The No. 4 beacon was very likely not visible from the channel over the half mile before the vessel left it. This very likely contributed.
- Clearing of the beacon's mangrove-screened sightline was deferred as unfunded for three works rounds, which very likely contributed.
- The beacon's conversion to solar halved its nominal range without any operator being told. This likely contributed.
- The aids to navigation code imposed no availability standard or notification duty on this channel. This likely contributed, though less firmly than the other factors.
- The change-of-master defect brief was verbal, and the morning's plotter freezes were not entered in the defect book. Each likely contributed.
- The alarm setting gave only 16 seconds' warning against a crash stop distance of 92 m, which extremely likely contributed.

The investigation could not determine whether new foreshore floodlighting behind the beacon affected the master's ability to see it. Tidal set-down and a steering or propulsion failure were both found very unlikely.

Two further matters did not contribute to the grounding but increased risk. The forward saloon lifejacket locker could not be reached from most of the seats it served, which very likely increased risk. The master's roster placed the darkness transit at the end of a long split duty late in a twelve-day tour, which likely increased risk.

No safety action has been recorded. [NOT ESTABLISHED: whether any safety recommendation is made, and if so its addressee and wording]

Factual information

All times are local (UTC + 10 hours).

History of the occurrence

On 07/07/2026 MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry operated by Tarrakin Ferries, was working a split roster between the town wharf at Kanooka Heads and Muntari Island. At the changeover of masters at the town wharf at about 13:05, the outgoing master told the relieving master that the chart plotter had frozen twice that morning and recovered by itself both times. The relieving master replied that it had done the same to him the month before. No entry of any kind was made in the wheelhouse defect book for 07/07/2026 before the occurrence.

Service TF18, the last of the day and one of the winter timetable's evening returns made in darkness since 01/06/2026, departed Muntari Island at 17:35 with 38 passengers, a master, a deckhand and a kiosk attendant on board. Sunset was at 17:02 and civil twilight ended at 17:29, so the channel transit was made in full darkness.

The vessel entered the Weerona Channel from the north at 18:03:58, abeam the No. 1 beacon. At 18:04:14 the master sent the deckhand to collect fares, leaving himself alone in the wheelhouse. At 18:05:30 the vessel was on the charted centreline abeam the No. 2 beacon, making 8.4 kt over the ground on a course of 195°.

At 18:07:06 the chart plotter display stopped responding: the chart picture remained with the vessel symbol stationary, and touch input was not accepted. The master tried the display twice, then continued at about 8 kt on visual reference, looking for the No. 4 starboard-hand beacon on the western side of the channel. Between about 18:07 and 18:08:50 the heading increased from 195° to 200° and was then held. The vessel crossed the western edge of the dredged channel at about 18:09:00. The master opened the wheelhouse window at 18:09:40 and sighted the No. 4 beacon at 18:10:02. In interview he said the beacon was dead ahead rather than on the starboard bow, and that he knew then that he was on the bank side. He also said he had eased the vessel five degrees to starboard while searching, without having decided to do so.

The echo sounder's shallow alarm sounded at 18:11:48, and both throttles were moved to astern together at 18:11:52. At 18:12:04 the vessel struck Boolarong Bank at 6.8 kt over the ground, 74 m west of the charted centreline and 54 m outside the dredged channel. It stopped within 11 m and came to rest heeled 4° to starboard on a falling tide.

The master stopped both engines at 18:12:31 and made no attempt to refloat, giving as his reasons the falling tide, the unknown bottom, the 38 people on board and the nine seconds of grinding. He addressed the passengers at 18:12:40 and broadcast a pan pan to Kanooka Heads Port Control on VHF channel 16 at 18:13:04; Port Control called out the port workboat. The deckhand sounded the No. 2 starboard void at 18:16, found water, and sounded it every ten minutes thereafter. All passengers were in lifejackets by about 18:22; issue in the forward saloon was delayed because the locker there is behind the kiosk servery counter. The workboat came alongside at 18:41 and the last passenger was ashore at 19:26.

The vessel was refloated on the rising tide at 00:20 on 08/07/2026 and towed to the town wharf, where a starboard hull breach and progressive flooding of the No. 2 void were confirmed. The Weerona Channel was closed to commercial traffic for 19 hours.

Injuries and damage

There were no serious injuries. One passenger reported a grazed shin, and two were treated at the wharf for distress.

The starboard hull bottom plating was set up over 4.1 m between frames 6 and 11, with a 180 mm split on the hull centreline in way of the No. 2 void. All three starboard propeller blades were bent aft, the starboard shaft was bent 1.4 mm over its length, the starboard rudder stock was distorted 3°, and the starboard skeg was torn away; it was recovered from the bank on 08/07/2026. The port hull, propulsion and steering were undamaged.

Recorded information

The shore AIS receiver log, engine monitoring recorder, plotter event log, echo sounder alarm log, bridge camera audio and VHF channel 12 and 16 recordings were recovered.

The AIS track shows the vessel within 3 m of the centreline until 18:07:06, then setting west at an average of 12 m per minute:

- 7 m west at 18:08:00
- 12 m west at 18:08:31
- 20 m west at 18:09:00
- 39 m west at 18:10:02
- 58 m west at 18:11:00, over Boolarong Bank
- 70 m west at 18:11:48

Rate of turn peaked at 2.1° per minute at 18:07:40. Speed over the ground fell from 8.4 kt to 7.6 kt between 18:07 and 18:11 with no throttle movement.

From 18:03 to 18:11:52 the engines ran at 1,780 and 1,784 rpm, within 12 rpm of each other, with no alarm, temperature or pressure excursion, or throttle movement. Steering hydraulic pressure stayed at 96 to 99 bar (normal band 90 to 105 bar), and rudder angle followed helm demand within 1.5°. At 18:11:52 both throttles moved to the astern detent together; astern thrust was developing but had not reached full at contact.

The plotter's event log recorded watchdog resets with cause code 0x51 (display process not responding) at 10:42 and 11:17 that morning. At 18:07:06 it recorded the display process not responding, with no reset. Power was removed at 18:16.

The echo sounder alarm sounded once on the voyage, 16 seconds before contact, during which the vessel covered 58 m.

Bridge audio records the master saying "Two ... good." at 18:05:33, "(unintelligible) frozen again." at 18:07:44, "Right. Where's four." at 18:08:31, "There's four. Keep her over." at 18:10:02 and "No no no." at 18:11:50. Sustained grinding and vibration followed the impact for approximately 9 seconds.

In the pan pan the master gave the position, persons on board and the absence of injuries and pollution, and stated: "I am not attempting to refloat." At 18:16:20 the deckhand estimated "Two, maybe three inches" in the void, which the master passed to Port Control at 18:17:05 as "approximately seventy five millimetres".

Equipment information

MV Tarrakin Star was certificated for 96 passengers, completed its annual survey on 11/03/2026 with no outstanding conditions, and was last slipped in 03/2025. None of the eleven defect book entries in the preceding 90 days concerned the chart plotter, GPS antenna, echo sounder, steering

gear, engines or propulsion controls. The maintenance system held no work order, fault report or correspondence about the chart plotter in 2026.

The plotter is a 12-inch multifunction display fed by a separate GPS antenna unit, running chart software revision 4.11 on firmware issued 03/2024. The echo sounder's shallow alarm had been set at 0.9 m under the keel since commissioning in 2019. The trials crash stop distance from 8 kt is 92 m.

The vessel carries 120 adult and 20 child lifejackets in four lockers. The forward saloon locker, holding 40 adult jackets, is behind a kiosk server counter fitted in a 2019 interior refit under a minor modification notice, which did not trigger a re-survey of the safety equipment arrangement. The arrangement approved at build in 2011 had an open bench there with the locker face clear. The same counter is fitted to the operator's two other 24 m vessels.

Meteorology

The master described the night as dark and clear, with a light westerly and flat water. There was no other traffic in the channel.

Personnel information

The master had 14 years' experience as a master, 11 on this route with Tarrakin Ferries, and approximately 4,100 Weerona Channel transits. He had held a certificate of competency as master less than 35 m since 2012; his last route check, on 14/02/2026, was satisfactory. He was on day nine of a twelve-day tour, working 05:40 to 08:40 and 13:10 to 19:20, and had slept 6 h 15 the night before. He described himself as tired, normal for a split, and neither fighting to stay awake nor slow.

The outgoing master had 9 years' experience as a master, 4 on this route, and approximately 1,400 channel transits. He had worked 05:40 to 13:05 on the day.

The deckhand had 7 months with the operator and was in her first winter season on the route. She held a certificate of safety training and a coxswain grade 2 qualification, and was in the saloons from about 18:05.

Organisational and management information

The operator's passage plan for the channel names a speed, a course and three beacons but says nothing about who is on the bridge; there was no night passage plan. Its appendix gives the No. 4 beacon's range as 6 M. OPS-3 is silent on a second person on the bridge in darkness. Both masters and the deckhand said the winter timetable change brought a new roster and nothing else.

Both masters said a plotter freeze that had cleared was not treated as a defect book matter. The outgoing master believed there had been three or four unrecorded freezes on the vessel in the past year. Asked what he would have done had it frozen inside the channel, he said he would have taken the way off and waited, though he knew of no rule requiring this.

Both masters said the No. 4 beacon had become harder to see at night over two years, and neither had reported it. The master said: "There's no form for the beacon looks dim." The operator's reporting system has a navigation hazard category, unused on this route in three years; no master interviewed knew it existed.

Port Authority records show the No. 4 beacon was converted from mains to a solar lantern in 11/2025, reducing nominal range from 6 to 3 nautical miles with the character, Fl G 2.5s, unchanged. The hydrographic service was notified on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator was told.

Work order WO-118, raised in 09/2023 for mangrove regrowth on Weerona Islet screening the beacon from the northern approach, was deferred as unfunded at the 2023, 2024 and 2025 works rounds. WO-176, raised in 03/2026, recorded that sightline as fully closed. A file note recorded that the islet, 240 m from the beacon, lay outside the 50 m scope of schedule 2.4 of the services agreement. The monthly check, made from the town wharf 2.1 nm south of the beacon where the islet is not in the line of sight, logged the beacon as operating every month from 01/2024. The Authority held no record of any operator enquiry about the channel's aids and had never provided the register to Tarrakin Ferries.

Both masters said the beacon had been harder to pick out since new car park floodlighting was installed behind it. The master said twice that he could not say whether the lighting was why he could not see it.

Tests and research

The steering rams, tiller arms, hydraulic system and autopilot were removed and function tested on 12/07/2026. All were serviceable, and the system passed a lock-to-lock timing test within the specified period.

The plotter was bench tested for 46 hours on 21/07/2026 with the same antenna and cabling; the fault did not recur and no defect was found. The contractor noted that cause code 0x51 is addressed by manufacturer firmware issued in 11/2024, which had not been applied; the operator held no subscription to the manufacturer's bulletins.

The grounding survey on 08/07/2026 placed the vessel 34 m west of the No. 4 beacon. The bed was 0.50 m below chart datum at the contact point, rising to 0.15 m below datum 30 m further west, a gradient of about 1 in 70. Furrows ran 11 m from first contact on a heading of 197°.

A range and visibility check from the workboat on 18/07/2026, on the inbound track at the same time of night, measured the lantern's luminous range at 2.6 nm. From the channel the light was not visible between 0.84 nm and 0.34 nm north of the beacon, because the sightline passes through the Weerona Islet canopy, measured at 5.8 m high against 2.1 m in a 2019 aerial survey. The lantern is 4.2 m above chart datum. On the night, the plotter froze at 0.68 nm, the master began looking for the beacon at 0.47 nm and called it at 0.25 nm.

A survey on 16/07/2026 found that the forward saloon locker cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter through a 0.6 m gap. The muster placard directs passengers to a locker not visible from the seats.

Analysis

External Influences

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

This subsection considers whether the standard governing the channel's aids to navigation shaped the conditions in which the No. 4 beacon could not be seen from the channel on the night. The organisational decisions that followed from that standard are argued in the next subsection. These were the solar conversion that halved the beacon's nominal range without any operator being told, and the three successive deferrals of the sightline clearing. They are referred to here only as what they were found to be.

It is virtually certain that the Weerona Channel's beacons were subject to no availability standard and no duty to inform users. This is documentary and not in dispute. Section 4.2 of the Waterways Aids to Navigation Code WAN-2 sets three duties for an aid on a declared shipping channel:

- to record its availability;
- to achieve "not less than 99.0 per cent measured annually";
- to "notify a change to the aid's character, position or range by notice to mariners before the change takes effect".

The Weerona Channel is not a declared shipping channel. Section 4.3 therefore applies, and it requires only that the responsible authority can demonstrate inspection "at intervals it has determined and recorded."

The services agreement between Kanooka Shire Council and the Kanooka Heads Port Authority sets out what that meant in practice. Its terms are:

- a daylight inspection once a quarter;
- a monthly confirmation of operation made from the town wharf, 2.1 nm south of the No. 4 beacon;
- vegetation management limited to 50 m from an aid;
- at S2.7, no requirement "to notify waterway users of a defect, a deferral or a change to an aid."

The Authority was not in breach of either instrument. Nothing in this subsection should be read as suggesting that it was.

How the standard could have influenced the outcome

Every mechanism that might have carried the beacon's condition to the people navigating by it stops at the boundary of this channel. These are a measured range, a recorded availability figure, and a notice to mariners of a change in range.

Within the scope the standard did set, the arrangements produced a consistent and reassuring picture. The monthly log recorded the No. 4 beacon as observed and operating in every month from 01/2024 to 07/2026, the last on 02/07/2026. That check was made from the one direction in which Weerona Islet is not in the line of sight. No defect was ever recorded against the beacon, and no notice about it was ever issued to a waterway user. The aids to navigation register, which held work order WO-118 and later WO-176 describing the screening, had never been provided to Tarrakin Ferries.

The effect of the missing notification duty is visible in the operator's own documents. The solar conversion in 11/2025 reduced the beacon's nominal range from 6 to 3 nautical miles. A

notification was lodged with the hydrographic service on 04/02/2026, but the chart correction was still pending on 07/07/2026 and no local notice was issued. On the night of the occurrence the appendix to Tarrakin Ferries' passage plan still recorded the No. 4 beacon at 6 M. Had section 4.2 applied, the Authority would have been required to notify the change before it took effect. That is the one mechanism in the evidence that would have put the altered beacon in front of the operator.

The 50 m vegetation limit had a similar effect. Weerona Islet lies 240 m from the beacon, so the Authority's file note placed its clearing outside the agreement's scope. The work competed for funding with boat ramps and pontoons rather than with aids to navigation.

The evidence against influence

The evidence against influence is substantial and was weighed.

First, nothing in the code prevented the Authority from notifying operators, trimming the mangroves, or measuring the light. The decisions not to do so rested on money and priority, not on anything the code required or permitted.

Second, an availability target of 99.0 per cent would not have detected this condition at all. The lantern was lit and working throughout. The difficulty was that it could not be seen from the channel between 0.84 nm and 0.34 nm north of the beacon, and no availability figure measures that.

On this reading the code's limited scope is background to the Authority's own choices rather than a cause of them.

How the matter was resolved

The investigation accepted that the standard's scope contributed, but it holds this conclusion with less confidence than most of the other influences in this report, and it is not presented as settled.

What tipped the balance was the notification duty. Its absence is the reason the range change never reached the operator, and the operator's passage plan still carried the superseded figure on the night.

The availability argument against influence was accepted as correct but narrower in reach. It shows that one part of section 4.2 would not have helped. It does not show that the section as a whole would have made no difference.

The factor is contributory, not causal. The Authority could have acted within the existing framework, and a broader standard would not by itself have removed the vegetation or altered what the master could see. What the standard's scope did was leave a narrow dredged channel used by a passenger ferry with beacons that nobody was required to measure, record, or report on.

It is likely that the absence of any availability standard or user-notification duty for aids to navigation on the Weerona Channel contributed to the grounding. It did so by leaving no required route by which the No. 4 beacon's reduced range and obscured sightline would reach those navigating by it. This constitutes a safety issue.

Organisational Influences

Winter timetable moved the last service into darkness without reopening the route risk assessment

The Weerona Channel was assessed, and operated, as a daylight route. RRA-07 issue 2, the current route risk assessment, named the hazards of the transit. These were Boolarong Bank to the west, a channel width of 40 m against a vessel beam of 8.2 m, and a "single unlit reach between the No. 2 and No. 4 beacons". It then imposed the restriction "daylight transits only". It went on to state that

"a night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan."

Change record CHG-2026-014 moved TF18 from 16:15 to 17:35 for the winter timetable beginning 01/06/2026. On 07/07/2026 the end of civil twilight was 17:29, so from the first day of the change the transit was made in darkness. The change was assessed as "timetable change only" and recorded "No new hazard identified". It answered "No" to whether a route risk assessment review was required. It did not refer to the daylight-only restriction. It was approved by the Operations Manager, one level below the approver the restriction names. RRA-07 was not reopened, no night passage plan was written, and the General Manager Operations' approval was neither sought nor given. TF18 then ran 31 times in darkness before the occurrence. The master made 26 of those transits.

The two documents are current controlled copies, and together they show that the change was made without the review the restriction required. What the Operations Manager understood the restriction to mean rests on an account that was given verbally and not in formal interview. For that reason, the existence of this factor is put as extremely likely rather than certain.

The question of influence is whether a review would have changed what happened on the night. A review of RRA-07 against a night operation would have had to address the unlit reach that the assessment already identified as a hazard. Under the terms of the restriction, it would also have required a specific passage plan before the first service ran. The vessel left the channel in that unlit reach, after losing its primary means of fixing. It is possible, however, that a review would have produced a plan saying no more than the daylight plan already said. The absence of a night passage plan and hold point is itself a separate factor, argued under Risk Controls.

It was likely that the timetable change, made without reopening the route risk assessment, contributed to the grounding by leaving the channel to be transited at night without the planning the assessment required. This was a contributory factor rather than a causal one.

Beacon converted to solar, nominal range halved, and no operator was told

In 11/2025 the No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal 3 nautical miles, against the 6 nautical miles of the lantern it replaced. The light's character, Fl G 2.5s, did not change. A notification was lodged with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator using the channel was told.

The operator's Weerona Channel passage plan, issue 2 and current, still listed "No. 4 (Fl G 2.5s, range 6 M)", and named that beacon as part of the secondary means of fixing. The luminous range measured from the port workboat on 18/07/2026 was 2.6 nm. At the time of the occurrence, the master was therefore working to a written plan that described a light more than twice the range of the one actually installed, and had been for about eight months. The conversion, the lantern ratings, the lodgement and the absence of any notice are all documentary, and the range was measured independently. It is virtually certain that this factor existed.

A weaker light is harder to acquire, and the master spent 91 seconds looking for the No. 4 beacon without finding it. However, almost all of that search took place within the stretch of channel where the light was screened entirely by vegetation. A screened light would not have become visible by being brighter. Whether the foreshore lighting behind the beacon affected its detection is an unresolved matter, discussed in its own subsection, and nothing from it is relied upon here.

It was likely that the reduced range of the No. 4 beacon, which had not been notified to users, contributed to the master's difficulty in acquiring it. This was a contributory factor.

Clearing of the beacon's sightline deferred unfunded for three works rounds

The Port Authority's register shows that the screening of the No. 4 beacon was known about well before the occurrence.

- **09/2023.** A work order identified mangrove regrowth on Weerona Islet screening the beacon from the northern approach. It recommended trimming to 2 m on a three-yearly cycle, at an estimated cost of \$18,500.
- **2023, 2024 and 2025 works rounds.** At each round, the work was deferred as unfunded and referred to the next.
- **03/2026.** A further work order recorded the sightline as fully closed on the northern approach.

The Authority's file note placed the islet 240 m from the beacon. This was outside the 50 m scope of schedule 2.4. As a result, the work competed for the general waterways vote, alongside boat ramps and pontoons, rather than for aids to navigation funding. The 2019 aerial survey gave the canopy height as 2.1 m. On 18/07/2026 the canopy measured 5.8 m, against a lantern height of 4.2 m above chart datum. The deferral is recorded in the Authority's own register and was confirmed by measurement on site. It is virtually certain that this factor existed.

The check on 18/07/2026 found that the light was not visible from the channel between 0.84 nm and 0.34 nm north of the beacon. The master began looking for the beacon at 0.47 nm and sighted it at 0.25 nm. In between, the vessel crossed the channel's western edge at 0.40 nm, inside the screened arc. During the period in which the master was relying on the beacon, it was not dim or defective but absent. The only reservation is that the master might not have picked the light up in that minute and a half even if it had been visible.

It was very likely that the unfunded deferral of the sightline clearing contributed to the vessel leaving the channel, by removing the No. 4 beacon from view over the stretch in which the master was looking for it. The evidence supports this as a contributory factor rather than a causal one, because it does not establish that the grounding would have been avoided had the light been visible.

Risk Controls

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

The transit was made under OPS-3 and its Appendix A, issue 2, which is the current issue. OPS-3 3.4 requires the channel transit to be "made on the charted centreline, monitored on the chart plotter". Appendix A names the plotter as the primary means of position fixing. It names the visual transit of the No. 4 beacon and the town wharf leading lights as the secondary means. It closes with the plan's only reference to darkness: "Night transits are not scheduled on this route." The operator confirmed that no night passage plan exists for the Weerona Channel. It is virtually certain that the plan contained no guidance on losing the plotter and no hold point or abort criterion of any kind.

The plan's secondary means was also weaker in practice than it appeared. The leading lights lie 2.1 nm beyond the No. 6 beacon and do not resolve as two lights until the vessel is inside the channel's southern third. Where the plotter was lost, the secondary means therefore depended entirely on the No. 4 beacon. Why that beacon could not be seen is argued elsewhere in this report. OPS-3 3.6 does require a reduction in speed and consideration of anchoring, but only "in reduced visibility". Visibility was not reduced that night, so the provision did not apply.

The master had 114 seconds between the display freezing and the vessel crossing the channel's western edge. The water was flat and there was no other traffic. A named hold point would have been available throughout that period. Both masters interviewed stated independently that the right course would have been to stop, and that nothing in the operator's procedures tells them to. One consideration weighs against the influence of the missing hold point. A master who did not think of stopping unprompted might not have applied a written rule under the same pressure. That consideration was weighed and does not displace the conclusion, but it limits how far the conclusion can go.

The absence of a hold point very likely influenced the decision to continue at eight knots rather than stop in the channel. Because the outcome still ran through that decision on the night, the evidence places this factor as contributory rather than causal. The absence of a night passage plan with a hold point for loss of the primary means of fixing very likely contributed to the transit continuing after the plotter froze.

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

OPS-9 9.1 requires immediate reporting only where a defect renders the vessel unseaworthy or affects a statutory item. On a class 1D vessel on this route, the chart plotter is not a statutory item, even though the passage plan names it as the primary means of position fixing. Under 9.2, all other defects are to be entered in the defect book "at the end of the shift in which they are found". For a fault that clears itself, that provides no effective trigger. Under 9.3, the outgoing master must brief the relieving master, but nothing requires the brief to be written down. Nothing requires the relieving master to confirm that a defect mentioned in it has been booked. Both masters described the same understanding independently: the book is for what is broken at handover, and what has come good is spoken about. The manual bears this out. It is virtually certain that the arrangement existed as described. It explains why the morning's two freezes were not written down, but it does not make the omission any less significant.

There are two readings of whether this arrangement influenced the grounding. The first reading looks beyond the single missed entry. The outgoing master believed the plotter had frozen three or four times on the vessel in the past year, and none of those occasions had been written up. The maintenance system held no plotter entry of any kind for 2026. The operator did not hold the manufacturer's bulletin that addressed cause code 0x51 and had no subscription to receive it. On this reading, an intermittent fault on the primary means of fixing never reached anyone ashore who could act. As a result, the firmware issued in 11/2024 was never applied, and the master met the fault that night with no reason to treat it as more than a nuisance.

The second reading concerns the day itself. An entry made at the 13:05 change of master would have been reviewed by the Operations Manager at the end of that week, not that afternoon, and TF18 would have sailed in any case. On that reading, the handover rule is a genuine finding about the fleet but not a cause of this grounding.

The investigation gave more weight to the first reading, because its argument rests on a year of unreported freezes rather than one afternoon's entry. The second reading is not dismissed. The arrangement likely contributed, as a contributory rather than causal factor, to the plotter fault going unrectified and being met in the channel without warning. That conclusion rests on the cumulative effect over the preceding year, not on any action that could have been taken on 07/07/2026.

Forward saloon lifejacket locker not reachable from most of the saloon it serves

The forward saloon locker holds 40 adult lifejackets and is mounted behind the kiosk serving counter. The survey of 16/07/2026 found that the locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter through a 0.6 m gap. When people are standing at the counter, that gap is the only route. The muster placard directs passengers to a locker they cannot see from their seats.

The arrangement approved at build in 2011 had an open bench in that position. The counter was fitted in a 2019 refit under a minor modification notice, which did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels. It is virtually certain that the condition existed.

It is unlikely that the condition played any part in the grounding or its outcome. All 38 passengers were in lifejackets within about eight minutes and ashore within 74 minutes. The vessel was aground, upright and stable. The deckhand's account shows the muster took somewhat longer than it should have.

The condition matters for a different reason. In a fire, a capsized risk, an evacuation into the water at night, or with a deckhand injured, a locker the passengers cannot reach without crew help would decide whether a saloon equips itself or waits. The underlying weakness is the modification route: an interior refit changed a safety equipment arrangement, and nothing in that route asked whether it had. The inaccessible locker, and the modification route that produced it, very likely represent an increased risk to passengers across the operator's class of 24 m vessels.

Local Conditions

Chart plotter display froze during the channel transit and did not recover

The plotter's internal event log recorded the display process as not responding at 18:07:06, under cause code 0x51, with no accompanying reset. The same code had been logged at 10:42 and 11:17 that morning, and on both occasions the unit recovered itself.

The bridge audio matched the log closely:

- "Oh, come on" at 18:07:11;
- sounds consistent with repeated screen taps at 18:07:26;
- "(unintelligible) frozen again" at 18:07:44.

The master's account was that the chart picture stayed on the screen, with the vessel symbol stationary and the display not accepting touch input.

The unit was bench tested for 46 hours on 21/07/2026 with the same antenna and cabling. The fault did not recur and no defect was found. The failure therefore rests on the software's own report of itself, corroborated by the audio and the master's account, rather than on an independently demonstrated fault. The contractor identified cause code 0x51 as known to the manufacturer and addressed by firmware issued in 11/2024, which had not been applied to this unit. That is consistent with the log. It is very likely that the display froze and did not recover.

While the plotter worked, the vessel held the charted centreline to within 3 m. It began to move west from 18:07:06 and crossed the channel's western edge at about 18:09:00. The master began looking for the No. 4 beacon 85 seconds after the freeze. It is extremely likely that the freeze influenced the occurrence, because it removed the primary means of position fixing that the passage plan named.

Two related matters are argued elsewhere. Why the morning's freezes went unrecorded is addressed under Risk Controls. Why the transit continued rather than stopping is addressed under Individual Actions.

The plotter freeze was very likely a causal factor: without it, the sequence that led the vessel out of the channel would not have begun.

No. 4 beacon not visible from the channel over the half mile before the vessel left it

The range and visibility check on 18/07/2026 was made from the port workboat, on the inbound track and at the same time of night. It found that the No. 4 beacon could not be seen at all from the channel between 0.84 nm and 0.34 nm north of it. Over that stretch the sightline passed through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. The canopy measured 5.8 m, against a lantern height of 4.2 m.

The check was made eleven days after the occurrence, and the canopy was not surveyed on the night itself. However, the screened arc was measured from the vessel's own track rather than modelled, and the bridge audio fits it:

- "Right. Where's four" at 18:08:31, with the vessel 0.47 nm north of the beacon;
- "There's four. Keep her over" at 18:10:02, at 0.25 nm and 91 seconds later.

Both masters interviewed raised the beacon without prompting. It is extremely likely that the beacon was screened over that stretch on the night.

The whole of the master's search therefore took place inside the screened arc. With the plotter frozen, the beacon was the secondary means of fixing named in the passage plan and the only one available. It was absent throughout the period in which the vessel was set west. By the time it came into view, the vessel was 39 m west of the centreline and outside the dredged channel. The master was not failing to look; there was nothing to see.

One qualification applies: the vessel had begun to set before the master started looking. It is very likely that the screening influenced the occurrence. The solar conversion and the deferred sightline clearance that produced this condition are addressed under Organisational Influences.

The screening of the No. 4 beacon was very likely a contributory factor: it did not set the vessel west, but it removed the only means by which the departure could have been detected and corrected before the bank.

New foreshore floodlighting behind the beacon reduced the master's ability to pick it up

The investigation examined whether foreshore floodlighting, commissioned on 22/05/2026, affected detection of the beacon. The lighting was commissioned ten days before the winter timetable began.

The case for an effect rested on the following:

- The survey of 18/07/2026 found 14 unshielded LED floodlights on 12 m columns, each presenting about 4,800 candela toward the channel.
- From 0.5 nm north of the beacon, the array lay behind it within 4 degrees of the same bearing and 1.5 degrees of the same elevation. It was continuously lit while the beacon showed a 0.4 second green flash every 2.5 seconds.
- The master described the beacon as lost against the light of the town and the new car park. The outgoing master independently described it as washed out.
- The Port Authority's aids to navigation officer was not consulted on the development, and no assessment against the channel's aids was made.

The case against was as follows:

- The only position at which the alignment was measured lay inside the screened arc, where the beacon would not have been visible against a dark background either.
- The master picked up the light once it opened past the islet, with the floodlights unchanged.
- No measurement exists of what could be seen on the night.
- Nothing establishes a detection threshold for a 0.4 second green flash against that background.

The evidence does not separate the effect of the floodlights from that of the canopy. This may not be separable while the mangroves remain, and settling it would need such a detection threshold established from positions where the sightline is clear. It is about as likely as not that the floodlighting affected detection. The matter remains unresolved, and nothing from it is carried into the conclusions.

Water level below prediction reduced the depth over Boolarong Bank

This was ruled out on the following figures:

- The Kanooka Heads gauge recorded 0.91 m at 18:12, against a prediction of 0.94 m.
- That left 1.41 m of water over the bed at the contact point.
- The vessel's static draught was 1.35 m, and its calculated squat was 0.18 m.
- The vessel was 54 m outside a channel dredged to 3.2 m.

Day nine of twelve, on a split duty across a thirteen hour forty spread

The master was on his ninth consecutive day of a 12-on, 2-off roster, on a split duty. The duty ran for 9 hours 10 minutes across a spread of 13 hours 40 minutes, inside both the operator's and the state's limits. He had recorded 6 hours 15 minutes of sleep the night before.

The recorded evidence gave no sign of degraded performance on the night:

- He held the centreline to within 3 m.
- He responded to the shallow alarm in 4 seconds, against a fleet figure of 6 to 9.
- His radio work after the grounding was correct at the first attempt.
- He decided not to refloat in about thirty seconds, and that decision was right.

Both masters independently said that no document prompted them to consider stopping. The outgoing master, who was not tired, described the same understanding of the handover and the same difficulty with the beacon. It is unlikely that the duty pattern influenced this occurrence.

It remains likely to be important as a standing exposure. It recurs on every darkness transit of the winter timetable, alongside a fatigue diary that the operator does not collect and a route with no relief master rostered. It is an other risk factor, not a contributory one.

Individual Actions

Plotter fault passed on verbally at the change of master and not entered in the defect book

The plotter's event log recorded two display freezes on the morning shift, at 10:42 and 11:17, under cause code 0x51, and the unit recovered by itself from both. At the change of master at about 13:05 at the town wharf, the outgoing master told the relieving master about both freezes. The relieving master replied that the unit had done the same to him the month before. Neither master entered the fault in the wheelhouse defect book or reported it to the Operations Manager. The defect book carried no entry of any kind for 07/07/2026 before the occurrence. The two masters'

accounts of the handover agreed in every particular, and both volunteered the omission before being asked about it. It is virtually certain that the fault was passed on verbally and not written down.

The omission was consistent with what OPS-9 required. Paragraph 9.3 required the outgoing master to "brief the relieving master on the state of the vessel, any defect, and anything else affecting the conduct of the service". Nothing required that brief to be written, and nothing required the relieving master to confirm that a defect mentioned in it had been entered. The chart plotter was not a statutory item, so an intermittent fault in it fell under 9.2. Under 9.2 the fault was due in the book only at the end of the shift in which it was found, with a weekly review ashore. The finding therefore concerns the practice that the procedure permitted, not the two individuals. The procedure itself is discussed under Risk Controls.

The consequence of the omission lay ashore. The maintenance system held no work order, fault report or correspondence about the plotter at any time in 2026. The operator held no subscription to the manufacturer's bulletins. The firmware issued in 11/2024, which addresses cause code 0x51, had therefore never been applied. An entry in the defect book would have brought the repeated freezes to the attention of those who decide on firmware and on withdrawing a vessel from service. The fault the master met at 18:07 was one the manufacturer had already addressed.

Two limits apply to this reasoning. First, a single entry made at about 13:05 would have been reviewed only weekly, so it would not by itself have kept TF18 from sailing that evening. Second, the unit did not reproduce the fault during a 46-hour bench test, and no defect was found in it.

It is likely that the absence of a written defect entry contributed to the plotter freezing during the transit. This was a contributory factor, not a causal one.

Transit continued at eight knots after the plotter froze, rather than stopping in the channel

The recorded track shows that speed was not reduced after the freeze at 18:07:06. The vessel held 8.4 kt to 18:08. The later fall to 7.6 kt by 18:11 occurred with no throttle movement on the engine recorder and is accounted for by drag from shoaling water. The bridge audio shows the master's attention turning to the beacon at 18:08:31, 85 seconds after the freeze, with "Right. Where's four." The audio contains no indication at any point that stopping was considered.

The master's own account is consistent with this. He did not consider stopping. He went directly to the visual transit because he regarded it as the backup. He said that, had he been asked beforehand, he would have said he would stop. The recorded track, the audio and his account all agree, and it is virtually certain that the transit continued at speed.

Stopping was available throughout. There was no other traffic in the channel, the water was flat and the wind was offshore at 8 kt. The vessel's trials crash-stop distance from 8 kt was 92 m. The vessel crossed the channel's western edge 114 seconds after the freeze. A stop at any point in that interval would have left it in the channel rather than on the bank.

No lookout was on the bridge. The deckhand had been sent to collect fares at 18:04:14. This was the normal practice on the route, and OPS-3 did not address it either way.

This action did not arise in isolation. The passage plan named the plotter as the primary means of fixing, but it contained no hold point and no guidance for losing the plotter. The beacon on which the visual backup depended was not visible over the stretch where the master was looking for it. Those conditions are analysed under Risk Controls and Local Conditions. Continuing at speed is the step at which they acted on the vessel.

It is extremely likely that continuing the transit at eight knots without a means of fixing position contributed to the vessel leaving the channel and to the late detection of shoaling. This was a contributory factor, not a causal one.

Engines stopped and no attempt made to refloat on a falling tide

Both engines were stopped at 18:12:31, 27 seconds after contact, and were not restarted. At 18:12:40, 36 seconds after the impact, the master addressed the passengers. At 18:13:04 his pan to Port Control stated "I am not attempting to refloat", and he warned that the tide was falling. He had the No. 2 starboard void sounded and reported ingress of approximately 75 mm at 18:17:05. The void was then sounded every ten minutes. All 38 passengers were given lifejackets.

The survey later confirmed a 180 mm split in the starboard plating in way of that void. Ingress reached approximately 260 mm by the time the vessel refloated at 00:20.

In interview, the master gave four reasons for not refloating:

- the tide was falling;
- the nature of the bottom was unknown;
- 38 people were on board;
- the vessel could not sink while it rested on the bank.

The breach was in the compartment the vessel was resting on. Backing off into 3.2 m of water, with the ingress not yet assessed and the passengers still aboard, would have put a flooding compartment into deep water at night.

It is virtually certain that the master stopped the engines and chose not to refloat. It is very likely that this decision reduced the risk to those on board.

Occurrence Events

Vessel left the dredged channel and tracked onto Boolarong Bank

The shore AIS track placed the vessel 1 m west of the charted centreline abeam the No. 2 beacon at 18:05:30. It held the centreline to within 3 m until 18:07:06, the time at which the plotter's event log recorded the display freeze.

From then the heading increased steadily from 195 to 200 degrees over about a minute and was then held. The vessel set west at an average of 12 m per minute. It reached the western edge of the 40 m channel at about 18:09:00, lay 39 m west of the centreline at 18:10:02 and was over Boolarong Bank from about 18:11:00. The surveyed grounding position was 74 m west of the centreline, 54 m outside the channel and 34 m west of the No. 4 beacon. That position, and the furrows in the bed running 11 m on a heading of 197 degrees, agree with the recorded track.

The shape of the departure matters as much as its extent. The recorded rate of turn peaked at 2.1 degrees per minute at 18:07:40 and was back to zero by 18:08:50. That is the pattern of a heading change applied and then held, not of a current or wind acting on the vessel. The wind was offshore at 8 kt, and there is no cross-set in the channel at that state of tide.

The departure began within a minute of the plotter freezing. When the master began looking for the No. 4 beacon at 18:08:31 ("Right. Where's four."), the vessel was already 12 m west of the centreline. When he sighted it at 18:10:02 ("There's four. Keep her over."), the vessel was outside the dredged channel.

Why the transit continued as it did after the freeze is argued under Individual Actions. The absence of the No. 4 beacon from view over that stretch is argued under Local Conditions. The possible

part played by new foreshore floodlighting behind the beacon is an unresolved matter, also discussed under Local Conditions, and nothing from it is carried into this conclusion.

The channel is dredged to 3.2 m, and outside it the vessel was over the bank. The grounding followed directly, and only, from the vessel being outside the dredged channel. It is virtually certain that the vessel left the channel and virtually certain that this led to the grounding. Had the vessel remained within the channel, it would not have grounded.

The vessel's departure from the dredged channel was a causal factor in the grounding.

Shoaling not detected until sixteen seconds before contact

The echo sounder's shallow alarm was set to 0.9 m under the keel. This was the value written into its setup at commissioning in 2019, and it had never been changed. The alarm sounded once on the voyage, at 18:11:48, when the vessel was 70 m west of the centreline. The bridge audio records the tone beginning at that second.

The response to it was prompt. Both throttles were moved to the astern detent together at 18:11:52, four seconds later. Contact followed at 18:12:04, 16 seconds after the alarm, at 6.8 kt over the ground. Astern thrust was developing but had not reached full. The alarm log, the bridge audio and the recorded speed agree to the second.

In those 16 seconds the vessel, at 7.1 kt, covered 58 m. The vessel's own trials figure for a crash stop from 8 kt is 92 m, so the warning given could not have been enough to stop the vessel.

Nor could the alarm have sounded appreciably earlier at that setting. Across the contact section the surveyed bank gradient is about 1 in 70, so the under-keel clearance fell to 0.9 m only a short distance before the bed rose to meet the hull.

The equipment did exactly what it was set to do. The shortfall lay in the combination of the setting and the speed at which the transit was being made. Had the setting given 40 seconds of warning, or had the speed been one at which 16 seconds was enough to stop, the vessel would not have touched. The circumstances in which the transit continued at eight knots after the plotter froze are argued under Individual Actions.

It is virtually certain that the shoaling was not detected until 16 seconds before contact. It is extremely likely that this late detection, against a stopping distance the vessel could not achieve in the time available, led to the grounding. Because an earlier warning or a lower speed would, on that evidence, have allowed the vessel to stop short of the bank, the late detection was a causal factor.

The detection of shoaling only 16 seconds before contact was a causal factor in the grounding.

Steering or propulsion failure caused the vessel to sheer off track

A steering or propulsion failure was considered, because a heading change followed by a steady set out of the channel can resemble a jammed rudder or a lost engine. It was found very unlikely and ruled out on two independent grounds:

- **Recorded data:** both engines ran within 12 rpm of each other with no alarm and no throttle movement, steering hydraulic pressure remained within its normal band, and rudder angle followed helm demand within 1.5 degrees throughout the transit.
- **Physical examination:** the complete steering system was found serviceable, with no sign of any pre-existing defect, when removed and function tested on 12/07/2026.

Conclusions

The investigation identified the following causal and contributory factors. They should not be read as apportioning blame or liability to any organisation or individual.

Findings

1. Waterways Aids to Navigation Code WAN-2 applied an availability target, a duty to record availability and a duty to notify changes by notice to mariners only to aids on a declared shipping channel. The Weerona Channel was not a declared shipping channel. None of those duties therefore applied to its beacons, and the services agreement did not require any user to be told of a defect, a deferral or a change. The Port Authority was not in breach of either instrument. (Safety issue)
2. Change CHG-2026-014 was approved by the Operations Manager on 22/04/2026. It moved service TF18 from 16:15 to 17:35, which placed it in darkness, and was assessed as a timetable change only. Route risk assessment RRA-07 was not reopened, although it restricted the channel to daylight transits and named the General Manager Operations as the approver of any night transit. That approval was neither sought nor given, no night passage plan was written, and TF18 ran 31 times in darkness before the occurrence. The Operations Manager's account of how he understood the restriction was given verbally and not in formal interview. (Safety issue)
3. The No. 4 beacon was converted to a solar lantern in 11/2025. The conversion reduced its nominal range from 6 to 3 nautical miles, and its luminous range was measured at 2.6 nm on 18/07/2026. No local notice to mariners was issued and no operator was told. The operator's passage plan in force on the night still recorded the beacon at 6 M. (Safety issue)
4. Mangrove regrowth on Weerona Islet was recorded as screening the No. 4 beacon from the northern approach under work order WO-118, raised in 09/2023. The clearing was deferred as unfunded at the 2023, 2024 and 2025 works rounds. By 03/2026, under WO-176, the sightline was recorded as fully closed. The islet lay 240 m from the beacon, outside the 50 m scope of schedule 2.4 of the services agreement. The monthly check was made from the town wharf, from which the screening could not be seen. (Safety issue)
5. There was no night passage plan for the Weerona Channel. The daylight plan named the chart plotter as the primary means of position fixing but gave no guidance on its loss, and named no hold point or abort point. The leading lights did not resolve until the vessel was inside the channel's southern third. At the point where the plotter was lost, the secondary means of fixing therefore consisted of the No. 4 beacon alone. (Safety issue)
6. Under OPS-9, the change-of-master defect brief did not have to be written down, and nothing required the relieving master to check that a defect mentioned in it had been entered. Non-statutory defects were due in the defect book only at the end of the shift in which they were found, which gave no trigger for an intermittent fault. The chart plotter was not a statutory item, although it was the primary means of position fixing. Three or four freezes over the preceding year were never written up, and firmware issued in 11/2024 that addressed the fault had never been applied. (Safety issue)
7. The forward saloon lifejacket locker could not be reached from 22 of the 34 forward saloon seats without passing behind the kiosk counter. The counter had been fitted in 2019 under a minor modification notice that did not trigger a re-survey of the safety equipment arrangement, and the same counter was fitted to the operator's two other 24 m vessels. This did not influence the grounding or its outcome, but it degraded a defence that would matter in a more serious evacuation. (Safety issue)

8. The chart plotter display very likely froze at 18:07:06 and did not recover. The unit's event log, the bridge audio and the master's account agreed. The fault did not recur during a 46-hour bench test and no physical defect was found. The cause code logged, 0x51, was one the manufacturer had addressed in firmware issued in 11/2024, and that firmware had not been applied to this unit.

9. The No. 4 beacon was very likely not visible from the channel between 0.84 nm and 0.34 nm north of it, because the sightline passed through mangrove canopy measured at 5.8 m against a lantern height of 4.2 m. The measurement was made eleven days after the occurrence. The plotter froze when the vessel was 0.68 nm north of the beacon, and the master began looking for it at 0.47 nm. The search therefore began inside the screened arc.

10. The investigation could not determine whether the new foreshore floodlighting behind the No. 4 beacon reduced the master's ability to pick it up. The evidence on this was about as likely as not either way.

- **For an effect:** both masters independently described the beacon as washed out against the town and the new car park. Fourteen unshielded floodlights commissioned on 22/05/2026 lay within 4 degrees of the beacon's bearing from 0.5 nm north.
- **Against an effect:** over the stretch where the master was looking, the beacon was screened by the canopy and would not have been visible against a dark background either. The master sighted it with the floodlights unchanged once it came clear of the islet. No measurement exists of what could be seen on the night.
- **Why it remains unresolved:** no detection threshold for a 0.4-second green flash against that background has been established, and such an assessment may not be possible while the canopy screens the light.

11. The water level over Boolarong Bank was very unlikely to have been below prediction to any material degree. The Port Authority's gauge recorded 0.91 m against a prediction of 0.94 m at 18:12. The vessel grounded because it was 54 m outside the dredged channel.

12. The master was on day nine of a twelve-day tour, on a split duty with a spread of 13 hours 40 minutes. This existed with virtual certainty and was inside every applicable limit. It was unlikely to have influenced the occurrence, because the recorded evidence showed no sign of fatigue on the night. It was likely to be a standing exposure that increased risk on every darkness transit of the winter timetable.

13. The two morning plotter freezes were passed on verbally at the change of master at about 13:05 and were not entered in the defect book. Both masters volunteered this omission. It was consistent with the practice that OPS-9 permitted.

14. After the plotter froze, the transit continued without any reduction of speed. There was no other traffic, the water was flat, and the vessel could have been stopped in 92 m. The master was alone in the wheelhouse, the deckhand having been sent to collect fares in line with normal practice on the route. OPS-3 did not address that practice.

15. The engines were stopped 27 seconds after contact, and no attempt was made to refloat the vessel on the falling tide with a breach in the compartment on which it was sitting. This was very likely to have reduced the risk to the 38 passengers.

16. From 18:07:06 the vessel's heading increased from 195 to 200 degrees and was then held, and the vessel set west out of the dredged channel. This was a heading change applied and held, not a current or wind effect. The vessel came to rest 74 m west of the centreline and 54 m outside the channel.

17. The echo sounder's shallow alarm was set at 0.9 m under the keel, the value used since commissioning in 2019. On a bank gradient of about 1 in 70, that setting gave 16 seconds' warning,

during which the vessel covered 58 m. The trials crash stop distance from 8 kt is 92 m. The equipment performed as it was set to do.

18. A steering or propulsion failure was very unlikely to have caused the vessel to leave the channel. The recorded engine and steering parameters were normal throughout. Every steering component was found serviceable on examination.

Causal factors

- It was virtually certain that the transit continued at eight knots after the plotter froze, rather than stopping in the channel, and very likely that this contributed to the grounding. The vessel crossed the channel's western edge 114 seconds after the freeze, and a stop at any point in that period would have left it in the channel rather than on the bank.
- The chart plotter display very likely froze during the channel transit and did not recover, and this very likely contributed to the grounding. The vessel held the centreline while the plotter worked and began to leave it within about a minute of the freeze.
- With virtual certainty, the vessel left the dredged channel and tracked onto Boolarong Bank, and it was virtually certain that this contributed to the grounding, which followed directly from the vessel being outside the dredged channel.

Contributory factors

- The winter timetable very likely moved the last service into darkness without reopening the route risk assessment, and this likely contributed to the occurrence. A review against a night operation would have reached the unlit reach between the No. 2 and No. 4 beacons and required a night passage plan, although such a review might have produced a plan that said no more than the daylight one. (Safety issue)
- There was, with virtual certainty, no night passage plan for the channel and no hold point if the primary means of fixing was lost. This very likely contributed, since a named hold point would have been available throughout the 114 seconds between the freeze and the channel edge. A master who did not think of stopping unprompted might not have applied a written rule under the same conditions, and the evidence did not settle whether this factor was causal or contributory. (Safety issue)
- That shoaling was not detected until sixteen seconds before contact was virtually certain, and it very likely contributed. At the 0.9 m setting the alarm gave 58 m of warning against a crash stop distance of 92 m.
- The change-of-master defect brief was verbal, with nothing checking that what was said was written down. This existed with virtual certainty and likely contributed, because repeated freezes never reached anyone ashore who could apply the firmware fix. An entry made at 13:05 would have been reviewed at the end of that week, and TF18 would have sailed either way, so the contribution is not put higher than likely. (Safety issue)
- Both masters passed the plotter fault on verbally at the change of master and did not enter it in the defect book. This was virtually certain and likely contributed, although a single entry on a weekly review cycle would not by itself have stopped TF18 that evening.
- Aids to navigation on this waterway carried, with virtual certainty, no availability standard and no duty to tell users anything, and this likely contributed. The notification duty would have been the mechanism that put the range change before the operator. Against that, nothing in the code prevented the Authority from notifying operators or clearing the sightline, and an availability target would not have detected a light that was lit but

screened. The balance of evidence for this contribution was less firm than for the other contributory factors. (Safety issue)

- The conversion of the No. 4 beacon to solar, which halved its nominal range, with no operator told, was virtually certain and likely contributed, since a weaker light is harder to acquire against a bright background. Over the stretch where the master was looking, however, the light was screened entirely, and a brighter light would not have been visible there either. (Safety issue)
- The No. 4 beacon was very likely not visible from the channel over the half mile before the vessel left it, and this very likely contributed. With the plotter frozen, the beacon was the only means of fixing available. The vessel may have begun to set west before the master started looking, and the evidence did not settle whether this factor was causal or contributory.
- Clearing of the beacon's sightline was deferred as unfunded for three works rounds, which was virtually certain, and this very likely contributed as the physical reason the light was absent from the screened arc. The master might have failed to pick the light up in that minute and a half even had it been unscreened. (Safety issue)

Safety action and recommendations

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

Under the Waterways Aids to Navigation Code WAN-2, the Weerona Channel's aids to navigation carried no availability standard and no duty to tell users anything about them. Section 4.2 of the code sets an availability target, a duty to record availability and a duty to notify changes by notice to mariners, but only for aids on a declared shipping channel. The Weerona Channel is not one. Under section 4.3, the responsible authority need only show an inspection interval it has set for itself. The services agreement set that interval, limited vegetation work to 50 m from the aid, and imposed no duty to tell any user of a defect, a deferral or a change. The Port Authority was not in breach of either instrument.

This matters because the code's threshold leaves a 40 m wide channel, used by a 96-passenger ferry, with beacons that nobody is required to measure, record or report on. It is likely that this contributed to the grounding. No required route existed by which the No. 4 beacon's halved nominal range and obscured sightline would reach the masters navigating by it. This contribution is less firmly established than the other contributory factors in this report. Neither the code nor the agreement prevented the Port Authority from notifying operators, clearing the sightline or measuring the light. An availability target would also not have detected a lantern that was lit but could not be seen.

No safety action has been recorded in response to this safety issue.

[NOT ESTABLISHED: whether a safety recommendation is to be made on this safety issue, and if so, its addressee and wording]

Beacon converted to solar, nominal range halved, and no operator was told

It was virtually certain that the No. 4 beacon had been operating at about half its former nominal range for eight months without any operator on the Weerona Channel being told. The beacon was converted to a solar lantern in 11/2025. Its nominal rating fell from 6 to 3 nautical miles, and a measurement from the port workboat on 18/07/2026 gave a luminous range of 2.6 nm. The hydrographic service had been notified, but the chart correction was still pending on the day of the grounding, and no local notice to mariners had been issued. Tarrakin Ferries' current passage plan appendix continued to record the beacon at 6 M.

The change mattered because of the gap it left between the plan and the light. The master was navigating to a written plan that described a light more than twice as strong as the one in front of him. That light also stood against a background of town and foreshore lighting added over the same period. The reduced range likely contributed to the beacon not being visible from the channel over the half mile before the vessel left it. It could be placed no higher than that for one reason: the stretch in which the master was looking for the light is also where the light is screened entirely, and a brighter light would not have been seen through that screening.

The issue arose because aids to navigation on this waterway carry no availability standard and no duty to inform their users.

No safety action has been recorded in response to this issue, and no safety recommendation is made in respect of it.

Clearing of the beacon's sightline deferred unfunded for three works rounds

Mangrove regrowth on Weerona Islet, 240 m from the No. 4 beacon, had been screening the light from the northern approach since at least 09/2023. In that month the Kanooka Heads Port Authority raised work order WO-118. It recorded the screening between about half a mile and a quarter mile from the beacon, recommended trimming to 2 m on a three-yearly cycle, and estimated the cost at \$18,500. The work was deferred at the 2023, 2024 and 2025 works rounds, each time with the entry 'defer, unfunded, refer next round'. In 03/2026 a further work order, WO-176, recorded the sightline as 'now fully closed on the northern approach'.

The Authority's file note against WO-176 explains why the work kept losing out. Because the islet lies outside the 50 m scope of schedule 2.4, the clearing had to compete for the general waterways vote rather than the aids to navigation vote. The canopy, recorded at 2.1 m in the 2019 aerial survey, measured 5.8 m on 18/07/2026. The lantern stands at 4.2 m.

That the clearing was deferred in this way is virtually certain. It very likely contributed to the grounding. On the night, the beacon was not dim or defective but absent from view: the screened arc runs from 0.84 nm to 0.34 nm north of the beacon, and the vessel left the channel inside it. The deferrals followed from the absence of any availability standard for aids to navigation on this waterway.

Safety action taken

No safety action has been recorded in response to this issue.

Safety recommendations

No safety recommendation is made in respect of this issue.

Winter timetable moved the last service into darkness without reopening the route risk assessment

The operator's winter timetable moved the last service of the day, TF18, into darkness, and the route risk assessment for the Weerona Channel was not reopened. RRA-07 restricts the channel to daylight transits. It requires the approval of the General Manager Operations and a specific passage plan for any night transit. The change that retimed TF18 was treated as a timetable change only and was approved one level below the approver the restriction names. Neither the General Manager Operations' approval nor a night passage plan was obtained.

It was likely that this contributed to the grounding. It left the channel to be transited at night without a night passage plan, and without a hold point in case the primary means of fixing was lost. It was a contributory factor rather than a causal one.

The issue matters because of how the change process worked. It asked whether the retiming introduced anything new, not whether the existing assessment still held for the operation as changed. As a result, a 96-passenger vessel was taken through a 40 m channel in darkness 31 times before the grounding, contrary to a restriction that remained in force throughout.

No safety action has been recorded in respect of this safety issue, and no safety recommendation is made.

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

The operator's passage plan for the Weerona Channel was written for the daylight transit. It stated that night transits were not scheduled on the route. The plan named the chart plotter as the primary means of position fixing and the visual transit of the No. 4 beacon with the town wharf leading lights as the secondary means. The leading lights do not resolve as two lights until the vessel is inside the channel's southern third. So at the point where the plotter was lost, the beacon was the whole of the secondary means.

It is virtually certain that no night passage plan existed for the channel. The plan also gave no guidance on losing the plotter and named no hold point or abort criterion. The only provision for slowing or anchoring, in OPS-3 3.6, applied in reduced visibility, and visibility was not reduced on the night. The absence of a hold point very likely contributed to the transit continuing at eight knots after the plotter froze. This was a contributory factor rather than a causal one.

The issue matters because this was the operator's own defence against the sequence that occurred, and it had never been written. The water was flat and there was no other traffic, so a hold point would have cost nothing. Both masters interviewed said independently that the right response was to stop, and that nothing in the operator's procedures told them to do so.

No safety action has been recorded in response to this safety issue.

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

The operator's Operations Manual OPS-9 required the outgoing master to brief the relieving master on the state of the vessel and any defect. It did not require that brief to be written down, and it required nothing of the relieving master afterwards. A defect that did not affect seaworthiness or a statutory item was due in the vessel defect book only at the end of the shift in which it was found, which gave no trigger at all for a fault that came and went. The chart plotter was the primary means of position fixing named in the passage plan, but it was not a statutory item. It is virtually certain that this arrangement existed. The manual text is documentary, and both masters independently described the same practice: the book was for what was broken at handover, and what had come good was spoken about.

It is likely that the arrangement contributed to the grounding. The outgoing master believed the plotter had frozen three or four times over the previous year, and none of those freezes reached anyone ashore who could act. As a result, a firmware fix issued in November 2024 was never applied, and the master met the fault on the night with no reason to treat it as more than a nuisance.

The contrary reading was also considered. An entry made at 13:05 would have been reviewed at the end of that week, and the vessel would have sailed TF18 either way. On that reading, the handover rule is a real issue for the fleet rather than a cause of this grounding. The arrangement explains why the fault went unbooked, but it does not excuse the omission.

No safety action has been recorded in response to this safety issue, and no safety recommendation is made.

Forward saloon lifejacket locker not reachable from most of the saloon it serves

The forward saloon lifejacket locker could not be reached from most of the saloon it served. The investigation did not set out to look for this. It came to light through the deckhand's account. While issuing lifejackets after the grounding, she had to ask three passengers to move away from the kiosk counter so that she could pass behind it, through a 0.6 m gap, to reach the locker.

A survey of the stowage on 16 July 2026 found the following:

- The locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the counter.
- With people standing at the counter, the inboard gap was the only route.
- The muster placard directed passengers to a locker they could not see from their seats.

The arrangement approved at build in 2011 had an open bench in that position, which left the locker face clear. The counter was fitted during a 2019 interior refit carried out under a minor modification notice. That notice did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.

The arrangement did not affect the grounding or its outcome. All 38 passengers were in lifejackets within about eight minutes. It is nonetheless very likely that it increased risk. In a fire, a capsized risk, a night evacuation into the water, or with a deckhand injured in the impact, the passengers could not have equipped themselves without crew help. The underlying gap lies in the modification route rather than the counter itself. An interior refit changed a safety equipment arrangement, and nothing in that route asked whether it had done so.

No safety action has been recorded in respect of this safety issue.

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